

[DC] Detection of an Isolated User's (Inter)Actions during an Augmented Reality Guided Procedure: an Application in Space Medicine

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ABSTRACT

With recent development of Augmented Reality (AR) and Artificial Intelligence (AI) technology, many new options have become available to overcome previously encountered problems (accurately detecting human activity, adaptive AR guidance, etc.). So when a collaborative scene between a "distant" user in an isolated location in need of assistance from a "local" user in a control centre is put into question, many different solutions can be considered. A solution could be to provide AR guidance for the isolated user with adjustable instructions from the local user who is immersed in a virtual representation of the isolated user's environment. This allows the local user to quickly understand the problem and current state of the procedure, enabling them to provide the following steps proactively or to rearrange the steps in the current procedure. During my thesis work, our aim is to study how to detect the interactions of an isolated user, in order to create a more detailed virtual environment, improving the quality and priority of the instructed actions provided by the local user. This research subject has become more feasible with recent developments in the fields of AR technology, Human Activity Recognition (HAR) for lightweight or non-intrusive equipment as well as Application Domain Models to represent the environment using semantic information.

Index Terms: Augmented reality, interactions, human activity recognition, application domain models.

1 THESIS SUMMARY

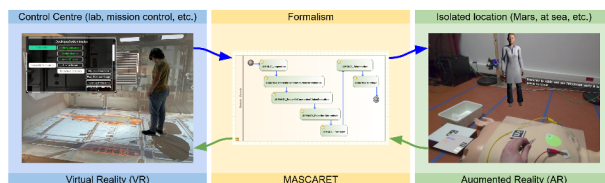


Figure 1: A local user in a control centre immersed in VR (on the left); a graph of the procedure in MASCARET [8]; a user in an isolated location assisted by AR instructions (on the right).

When an isolated user (e.g., sailor, astronaut, soldier, etc.) is in need of assistance during a critical situation (technical or medical), they may need to collaborate with a potentially far away expert in a control centre possessing crucial knowledge. To enable this distant collaboration, we propose exploiting the advantages of both Augmented Reality (AR) and Virtual Reality (VR). The isolated user in AR will be guided by AR instructions (e.g., with a head mounted display) while the expert will be immersed in VR within a virtual recreation of the remote scene (e.g., using a CAVE) promoting their decision-making based on what they visualise in VR. Basing our

work on a model for Intelligent Virtual Environments [8], the expert would be able to follow the progression of the tasks performed by the distant user as well as their state (started, finished without error, with error, etc.). To allow for this monitoring of states, the system must be able to detect and recognise the interactions and therefore actions of the isolated user. This information would allow the expert to follow the progression but also allow them to change steps in the procedure if an error or mistake occurs. Each procedure modification would then become visible to the isolated user in AR. My thesis work mainly tackles the automatic detection of the isolated user's current actions. However it is also important to note that an "isolated environment" may include not only a geographical difference (at sea, space, desert, etc.) but also latency due to transmission delay. My work therefore seeks to find answers to two major research questions: 1) *How to use HAR algorithms on continuous multimodal data streams and semantic elements of the activity model to detect specific interactions in AR?* 2) *What forms and combinations of input are available for our application domain in AR for Human Activity Recognition (HAR)?*

To answer these research questions in accordance with the limitations, we envisage the implementation of an environmental knowledge and human activity representation meta-model (e.g., MASCARET,...) which contains the semantic and contextual information of the scene, allowing for effective exchange of information despite the potential latency. This representation without additional input is however restricted to the prior knowledge of the physical environment. To keep the representation accurate to the scene, we plan to apply deep-learning techniques for HAR, in combination with the encoding of any spatio-temporal data if needed, to facilitate and improve the identification of the actions performed. This data will create a more in-depth virtual representation of the environment for the expert. These algorithms however require prior labelled examples and information about each detectable action in order to classify them. This notion can also be enhanced with the application of IoT (Internet of Things) for example with sensors mounted on objects present in the environment. This could provide information about their motion in space, enabling inference of the action in progress for the knowledge representation model. The problem posed in our research questions also includes how to best apply and combine these inputs and methods in order to display the information needed. The whole will allow for the design of a formalism representing the context and actions of the isolated user.

2 RELATED WORK

For Bowman *et al.* [2], there exists three "universal" interaction tasks for supporting immersive Virtual Environments (VE): *navigation*, *selection* and *manipulation*. While *navigation* may be less interesting outside of the world of VR as it is not intrinsically linked with the physical world, in AR, *selection* and *manipulation* can make up the majority of interactions. On the other hand, System control is also an important complex task which accounts for the user's input into the system. However, we shall consider it as part of Manipulation. In general terms, the inputs used to perform these universal interactions consist of Unimodal interactions (gesture, speech, object, etc.) which are commonly categorised into three groups: visual-based, sensor-based and audio-based [10]. These

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inputs allow for the implementation of interaction techniques such as 3D interactions, tangible interactions, spatial awareness interactions and multimodal interactions [7]. Multimodal interactions specifically are of high interest as their inputs can be complimentary and synergistic, for example speech can act as a good quantitative input while gesture as a good qualitative input [3]. Since high quality physical interactions are difficult to implement, it is common to use indirect speech or gesture interactions instead [1]. For our work, it is important to note that each action is composed of one or multiple interactions. However, our final objective is to detect and identify these actions. For this we make use of a field called Human Activity Recognition (HAR) which is very active due to advances in IoT making data more and more accessible in everyday life [9]. Applications in this field can be categorised into two groups: *sensor-based* and *video-based* [12]. The first involving the analysis of data from sensors such as Inertial Measurement Units and the second involving extracting features from videos or images containing human movement with tools such as a Kinect. This data often requires real time processing and Convolutional Neural Networks are well suited for feature extraction in this way [6]. It is also relevant that these actions can be easily identified for both users despite our imposed limitations. Using the aforementioned HAR techniques, extracted features can be associated with specific steps within a procedure or task. This means that the system is capable of knowing the current progression of the procedure and can update the VE accordingly. For this there exists Application Domain Models which allow for the creation of VEs from contextual and semantic information [8]. In application terms, our work is in line with research already carried out where communication or collaboration between an operator and an expert can be essential. For example, this has namely been tested for space procedures, for example MobIPV [5] (mobile procedure viewer) for establishing a link between two distant users for maintenance procedures and AR Coach [4] which provides medical assistance for a relevant POCUS-guided procedure relayed by a virtual coach. These three main concepts brought up in this section (Interaction, HAR and Application Domain Models) as well as their associated works currently serve as a basis and entry point for the work of this thesis.

3 PREVIOUS WORK & MOTIVATIONS FOR DC PARTICIPATION

In the same context as my current thesis work, which began in October 2024, I had the opportunity during my end-of-studies internship to tackle the issue of visualising information provided during AR guidance within the context of an isolated medical procedure. I first carried out a bibliographic study exploring AR applications in micro-gravity environments for space operations. I then worked on the comparison of two visualisation formats: Embedded information and "Projected Situated" information. By conducting an experiment with 27 workplace safety/first-aid qualified following EN ISO 14971 regulation participants, we found positive evidence towards the application of embedded visualisation in terms of cognitive load and acceptability compared to our "Situated Projected" visualisation. In contrast, our "Situated Projected" visualisation showed overall more reliability for high precision areas. This work was the subject of a scientific paper, of which I was the primary author [11], presented at the ACM VRST'24 conference. This preliminary work contributed to how we will study and design the future VE planned in this thesis. All of the previous work I have carried out in the field has nurtured a great interest in other works surrounding both AR and VR for me. Attending the Doctoral Consortium will allow me to meet fellow PhD students as well as international experts working on similar projects, but also to have additional visions and perspectives on my future work and to broaden my knowledge on current works in different fields applying these technologies.

4 EXPECTED CONTRIBUTIONS AND FUTURE WORK

My work aims to contribute: 1) *The analysis and testing of possible forms of AR interaction inputs and actions for various contexts.* 2) *The design and implementation of a system capable of detecting a user's actions in AR using a Human Activity Recognition model and the semantic information in the environment during a procedure.* 3) *A user study demonstrating the detection and recognition of the actions performed by a user.* Since my current work involves identifying methods to detect AR interactions, I propose the following topics for the doctoral consortium: 1) *What sort or combination of inputs would allow for the most in-depth yet reliable information for HAR?* 2) *In what way or how do IoT or physical/tangible objects, which an isolated user could make use of to interact with virtual information, allow for the detection of the user's actions?*

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